

Early Gradient Spectra Predict Useful Low-Rank Adaptation

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Abstract

LoRA and related PEFT methods expose a nominal rank, but useful rank depends on the task spectrum seen through the model’s activations. For a frozen linear map with activation covariance C and task update Δ_* , we show that the best rank- r adapter is the truncated SVD of $M_* = \Delta_* C^{1/2}$ and has excess risk $\frac{1}{2} \sum_{i>r} \sigma_i^2(M_*)$. The activation-whitened population gradient at the frozen model has exactly this spectrum, making early-gradient spectra candidates for predicting useful adapter rank before training. In five-seed matrix simulations, activation-whitened gradient effective rank predicts a prespecified near-best useful-rank target with mean $R^2 = 0.777$ and mean Spearman correlation 0.852 in a hard-knee regime; the corresponding sample-limited values are 0.687 and 0.786. We then test the stronger extrapolation from rank prediction to cross-module rank allocation in a prespecified synthetic-transformer study: two task families, five independent base-model seeds per task, three adaptation replicates, common random numbers, exact-cost uniform comparators, and fixed update scale as the primary condition. For soft spectral dimension versus exact-cost uniform allocation, the equal-task-weighted final-validation-loss difference (candidate minus baseline) is +0.0146 with a 95% task-stratified bootstrap interval $[-0.0078, 0.0352]$ and exact two-sided sign-flip $p = 0.2402$; the candidate wins 3 of 10 runs and neither task-family mean. Thus the confirmatory transformer experiment does not support a general allocation advantage. Exploratory standard- α/r scaling changes the direction but remains uncertain, exposing rank-scale sensitivity. A small GPT-2/Wikitext-2 experiment is retained only as a descriptive real-model check. The supported conclusion is narrower: early-gradient spectra predict useful rank in the reduced-rank model and controlled matrix simulations, while a universal module allocator is not established.

Scope and claim boundary. Most evidence is theorem-driven and synthetic. The central empirical claim is useful-rank prediction in matrix simulations, not universal rank allocation in transformers. The corrected synthetic-transformer experiment is a controlled boundary test and yields a null confirmatory result under its primary fixed-scale condition. The GPT-2/Wikitext-2 study is small and descriptive. We do not claim nonlinear transformer convergence, broad task generalization, or dominance of any spectral allocator over exact-cost uniform or gradient-based baselines.

1 Introduction

Low-rank adaptation methods such as LoRA make rank a visible and tunable resource. A rank- r LoRA adapter learns

$$\Delta W = \frac{\alpha}{r} B A, \quad (1)$$

with $B \in \mathbb{R}^{d_{\text{out}} \times r}$ and $A \in \mathbb{R}^{r \times d_{\text{in}}}$. In practice, however, nominal rank is an incomplete description of adaptation capacity. Rank interacts with scale, optimization time, activation geometry, finite-sample noise, and the module on which the adapter is placed. A rank that is algebraically available may not be functionally useful, and a small budget spent on the right module can outperform a larger budget spread uniformly.

We separate *nominal rank* from *useful rank*. Nominal rank is the algebraic rank supplied to the adapter. Useful rank is the smallest rank satisfying a practical criterion: near-best validation loss, a fixed fraction of attainable improvement, or a rank-penalized tradeoff. Our central observation is that useful rank is a spectral quantity in function space. If C is the activation covariance and Δ_* is the task update, the relevant operator is

$$M_* = \Delta_* C^{1/2}. \quad (2)$$

This activation weighting discounts update directions that the model rarely uses.

We prove two exact population statements in a reduced-rank adaptation model. First, the best rank- r adapter is obtained by truncating the singular spectrum of M_* . Second, the early full gradient at the frozen model, after activation whitening, equals $-M_*$. Therefore, the useful-rank spectrum is visible before training an adapter. We validate this prediction in controlled matrix simulations. We then test a stronger, non-equivalent hypothesis: whether a scalar spectral score can allocate a fixed LoRA parameter budget across transformer modules. A prespecified ten-run study with exact-cost comparators and common random numbers does not show a confirmatory advantage for the primary spectral allocator, and sitewise score-rank relationships reverse sign across the two task families. This negative bridge result is informative because it separates theorem-supported rank prediction from a practical allocation heuristic that requires further structure.

Contributions.

1. We give an exact reduced-rank adaptation theorem: rank- r excess risk is the tail singular energy of $\Delta_* C^{1/2}$.

2. We show that the activation-whitened population early gradient has the same spectrum as this task operator.
3. We define useful-rank targets—near-best, recovery, and penalized rank—that avoid the degeneracy of unconstrained best rank.
4. We validate useful-rank prediction in multi-seed spiked matrix simulations, where gradient effective rank strongly predicts practical rank targets.
5. We conduct a prespecified synthetic-transformer allocation study with ten independent task-seed runs, exact-cost baselines, common random numbers, three adaptation replicates, and explicit scaling controls; its null confirmatory result identifies a boundary between rank prediction and module allocation.
6. We retain a small GPT-2/Wikitext-2 experiment as a descriptive real-model check and state its module-family and stochastic-control limitations explicitly.

2 Related Work

LoRA and PEFT. LoRA freezes pretrained weights and trains low-rank increments, giving strong fine-tuning performance with far fewer trainable parameters [1]. QLoRA backpropagates through quantized frozen models into LoRA adapters, making LoRA-style adaptation central to memory-efficient fine-tuning [2]. Our focus is orthogonal: how much low-rank capacity is useful, and how should it be allocated?

Adaptive rank and scaling. AdaLoRA explicitly reallocates rank budget across weight matrices using importance scores [3]. Rank-stabilized LoRA changes the standard α/r scaling to mitigate high-rank learning issues [4]. DoRA decomposes weights into magnitude and direction and applies low-rank updates to the direction component [5]. These works motivate rank allocation and rank-scale disentanglement. We contribute a pre-training spectral predictor of rank demand.

Gradient subspaces and spectral perturbation. LoRA-One studies one-step full-gradient subspaces and shows that singular directions of the gradient can guide low-rank adaptation [6]. We ask a complementary question: not only which subspace, but how many directions. Mathematically, we build on Eckart–Young–Mirsky low-rank approximation [8, 9], classical perturbation theory [10], and spiked random-matrix transitions [13]. Pythia is a natural future open-model validation suite because it provides controlled checkpoints across scale and training time [7]; we do not use it in this paper.

3 Theory

Let

$$y = (W_0 + \Delta_\star)x + \xi, \quad (3)$$

$$\mathbb{E}[x] = 0, \quad \mathbb{E}[xx^\top] = C \succ 0, \quad \mathbb{E}[\xi | x] = 0. \quad (4)$$

A rank-constrained adapter predicts $(W_0 + \Delta)x$ with $\text{rank}(\Delta) \leq r$, and

$$\mathcal{L}(\Delta) = \frac{1}{2} \mathbb{E} \|y - (W_0 + \Delta)x\|_2^2. \quad (5)$$

Define $M_\star = \Delta_\star C^{1/2}$.

Theorem 3.1 (Population reduced-rank adaptation). *Let $M_\star = U\Sigma V^\top$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots$. The best rank- r adapter is*

$$\Delta_r^\star = U_r \Sigma_r V_r^\top C^{-1/2}, \quad (6)$$

and its excess risk is

$$\mathcal{L}(\Delta_r^\star) - \mathcal{L}(\Delta_\star) = \frac{1}{2} \sum_{i>r} \sigma_i^2(M_\star). \quad (7)$$

The smallest rank recovering fraction ρ of the attainable improvement is

$$r_{\text{rec}}(\rho) = \min \left\{ r : \sum_{i \leq r} \sigma_i^2(M_\star) \geq \rho \sum_i \sigma_i^2(M_\star) \right\}. \quad (8)$$

Proposition 3.2 (Early-gradient spectrum). *Let $G_0 = \nabla_\Delta \mathcal{L}(0)$. Then*

$$G_0 = -\Delta_\star C, \quad -G_0 C^{-1/2} = \Delta_\star C^{1/2} = M_\star. \quad (9)$$

Thus the activation-whitened early gradient has the spectrum that determines useful rank.

Predictors. We compare hard detectable rank, stable rank, effective rank, and soft spectral dimensions. The strongest replicated predictor is effective rank:

$$r_{\text{eff}}(A) = \exp \left(- \sum_i p_i(A) \log p_i(A) \right), \quad p_i(A) = \frac{\sigma_i^2(A)}{\|A\|_F^2}. \quad (10)$$

A threshold-aware continuous alternative is

$$d_{\phi, \tau}(A) = \sum_i \phi \left(\frac{\sigma_i^2(A)}{\tau^2} \right). \quad (11)$$

Hard detectable rank corresponds to the discontinuous gate $\phi(t) = \mathbf{1}\{t > 1\}$.

Finite samples. With empirical gradient $\widehat{G}_0$, ideal whitening gives $\widehat{M} = -\widehat{G}_0 C^{-1/2} = M_\star + E$. If $\|E\|_{\text{op}} \leq \varepsilon$, Weyl’s inequality controls singular values. Hard thresholding is stable only with a gap around the edge; effective rank is continuous under relative Frobenius perturbation. In practice we use regularized empirical whitening:

$$\widehat{M}_\lambda = -\widehat{G}_0 (\widehat{C} + \lambda I)^{-1/2}. \quad (12)$$

4 Experimental Setup

The matrix and small-transformer experiments are synthetic and require no external datasets. The final GPT-2/Wikitext-2 validation uses local pretrained-model and dataset files to test whether the diagnostic survives one real pretrained-transformer setting. The goal is to test the rank-prediction mechanism under controlled task spectra, sample noise, rank grids, scale choices, and a small real-model check. Table 1 summarizes the experimental regimes.

Synthetic regression. Tasks use $y = (W_0 + \Delta_\star)x + \xi$. Default dimensions are $d_{in} = d_{out} = 128$ with power-law activation covariance. The main task-rank-8 spike values are

$$(2.8, 2.2, 1.8, 1.45, 1.1, 0.85, 0.65, 0.5). \quad (13)$$

LoRA factors are trained by gradient descent with clipping. Detection thresholds use a permutation null at quantile 0.995.

Useful-rank targets. Let $\mathcal{R}$ be the tested nominal-rank grid. For validation curve $V(r)$, define $V_0 = V(0)$, $V_{\min} = \min_{r \in \mathcal{R}} V(r)$, $\Delta V = V_0 - V_{\min}$, and $Q(r) = (V_0 - V(r))/\Delta V$. When $\Delta V > 0$, we report

$$\begin{aligned} r_{\text{near}}(\gamma) &= \min\{r \in \mathcal{R} : V(r) \leq V_{\min} + \gamma\Delta V\}, \\ r_{\text{rec}}(\rho) &= \min\{r \in \mathcal{R} : Q(r) \geq \rho\}, \\ r_{\text{pen}}(\lambda) &= \operatorname{argmin}_{r \in \mathcal{R}} \{V(r) + \lambda\Delta V r / r_{\max}\}. \end{aligned}$$

Ties are resolved in favor of the smallest nominal rank. Curves with $\Delta V \leq 0$ have undefined useful-rank targets and are excluded from target fits. The simulation oracle loss is retained only as a diagnostic and never enters $V_{\min}$, ΔV , a threshold, or a penalty. The unconstrained observed-best rank is likewise diagnostic; in layerwise experiments it is often the largest rank tested.

5 Results

5.1 Detectability and nominal-rank failures

The BBP-style simulation confirms the expected detectability transition (Figure 1, left). In rank-1 Gaussian noise with edge near 2.0, planted/empirical overlap is near zero for small spikes and rises after spectral separation: mean overlap is about 0.008 at $\theta = 0.1$, 0.288 at $\theta = 1.2$, 0.540 at $\theta = 1.5$, and 0.893 at $\theta = 3.0$.

The rank-sweep panels in Figure 1 show why nominal rank alone is insufficient. A cleaned fixed- α LoRA sweep with $\alpha = 16$, learning rate 0.2, 1200 steps, ranks 1–64, and 8 repetitions shows a U-shaped curve. Mean validation loss is 0.1011 at rank 1, 0.0211 at rank 4, 0.0058 at rank 8, 0.0174 at rank 16, and 0.1303 at rank 64. Robust fits show that final adapter stable/effective/detectable rank explain validation loss far better than nominal rank.

5.2 Rank and scale are confounded

Alpha controls both optimization and drift. In the alpha stress run, validation loss improves up to moderate α and then degrades mildly, while the output-drift proxy increases. Large α did not robustly create separate “intruder” directions in this synthetic setting; the safer interpretation is amplification and drift. A rank-alpha grid further shows that nominal rank alone is weak: on the clean subset with validation loss < 1 , adapter stable rank reaches $R^2 \approx 0.810$ for log validation loss, adapter detectable rank plus alpha reaches $R^2 \approx 0.778$, adapter effective rank reaches $R^2 \approx 0.740$, alpha alone reaches $R^2 \approx 0.552$, and nominal rank alone reaches only $R^2 \approx 0.040$.

5.3 Signed conflict is secondary but informative

The original unsigned merge-overlap experiment was weak: spectral interference explained only about $R^2 = 0.051$ for merge degradation and planted overlap about $R^2 = 0.046$. A corrected signed-conflict experiment gave a clearer story: the signed task inner product explains merge degradation with $R^2 = 0.9995$, while a conflict score gives $R^2 = 0.747$ and unsigned overlap alone gives $R^2 = 0.085$. This is a secondary result; it supports using signed spectral geometry rather than unsigned overlap for task-vector interference.

5.4 Stage4: early-gradient effective rank predicts useful rank

The main matrix-simulation evidence is the Stage4 replication: five seeds per condition and 48 synthetic layers per seed. For each layer, we compute activation-whitened early-gradient spectral statistics, run rank sweeps over $\{1, 2, 4, 8, 16, 32\}$, and evaluate useful-rank targets. Before the corrected rerun, we designate the hard-knee $r_{\text{near}}(0.1)$ versus activation-whitened gradient effective rank comparison as the primary confirmatory pair; its sample-limited counterpart is a prespecified robustness check. Other displayed targets and all alternative predictors are secondary or exploratory. Figure 2 and Table 2 report the replicated activation-whitened effective-rank predictor.

Table 2: Stage4 activation-whitened effective-rank results. Mean over five seeds; SEM in parentheses.

Condition	Target	Mean R^2	Mean ρ
Hard-knee	near-best 0.1 gap	0.777 (0.041)	0.852 (0.017)
Hard-knee	near-best 0.2 gap	0.821 (0.020)	0.868 (0.013)
Hard-knee	70% recovery	0.808 (0.035)	0.872 (0.019)
Hard-knee	80% recovery	0.821 (0.020)	0.868 (0.013)
Hard-knee	penalty 0.2	0.707 (0.039)	0.804 (0.030)
Hard-knee	penalty 0.3	0.703 (0.034)	0.811 (0.023)
Sample-limited	near-best 0.1 gap	0.687 (0.038)	0.786 (0.026)
Sample-limited	near-best 0.2 gap	0.635 (0.042)	0.766 (0.030)
Sample-limited	70% recovery	0.573 (0.046)	0.731 (0.039)
Sample-limited	80% recovery	0.635 (0.042)	0.766 (0.030)
Sample-limited	penalty 0.2	0.614 (0.055)	0.740 (0.040)
Sample-limited	penalty 0.3	0.591 (0.036)	0.720 (0.019)

Table 1: Experimental settings. Stage4 is the primary evidence for early-gradient useful-rank prediction.

Experiment	Regime	Sweep	Purpose
BBP	$m = n = 128$, rank-1 spike	spike strength θ	verify spectral detectability transition
LoRA rank	$d_{in} = d_{out} = 128$, task rank 8	ranks 1–64	test nominal rank and final adapter spectra
Alpha / rank-alpha Merge/conflict	noisy or rank-8 tasks paired task vectors	α and rank grids overlap/sign sweeps	separate rank, scale, and instability test signed versus unsigned interference
Layerwise hard-knee	48 synthetic layers	ranks 1–32	test early-gradient useful-rank prediction
Layerwise sample-limited	fewer samples and more noise	ranks 1–32	test finite-sample degradation
Stage4 replication	5 seeds $\times$ 2 regimes	48 layers per seed	multi-seed validation
Synthetic trans-former	2 tasks $\times$ 5 seeds; 3 adaptation replicates	ranks 0–16, 3 scale rules, exact-cost budgets	test whether spectral rank demand transfers to module allocation
GPT-2/Wikitext-2	small local pretrained-model check	fixed-scale restricted module families	descriptive transfer and boundary test

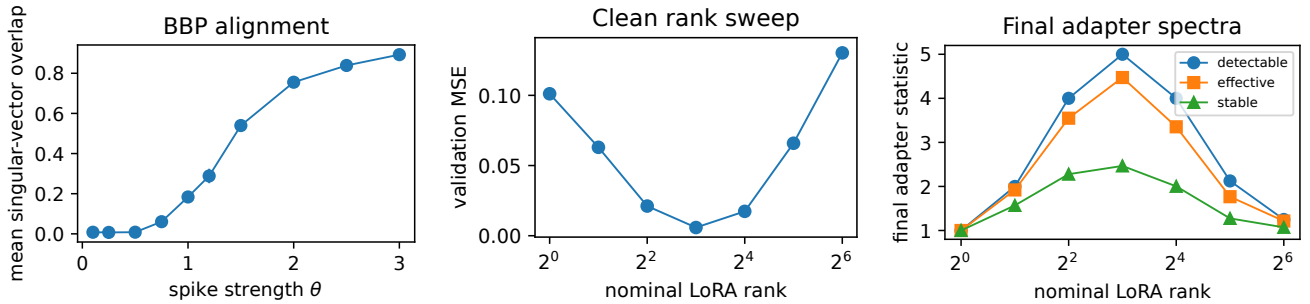


Figure 1: Left: BBP detectability transition. Middle: a clean fixed- α LoRA rank sweep is U-shaped, so nominal rank is not the explanatory variable. Right: final adapter spectral statistics peak near the useful-rank range.

The hard-knee regime uses $n_{train} = 1024$, label noise 0.05, and weak-factor range 0.01–0.08. The sample-limited regime uses $n_{train} = 384$, label noise 0.12, and weak-factor range 0.05–0.25. The unconstrained best rank is degenerate: rank 32 is best for almost every layer. Useful-rank targets are nondegenerate and predictable.

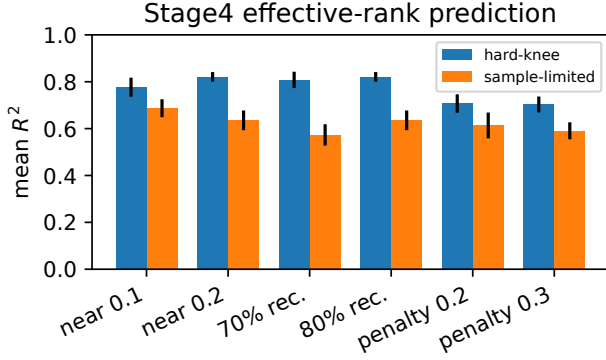


Figure 2: Stage4 replication. Bars show mean R^2 over 5 seeds for activation-whitened gradient effective rank predicting log2 useful-rank targets. Hard-knee spectra are strongly predicted; sample-limited spectra are weaker but still informative.

6 Synthetic-Transformer Allocation Test

We next test a stronger claim than the matrix theorem: whether modulewise early-gradient spectra can allocate a limited LoRA budget inside a nonlinear transformer. The study was frozen before the full run and is conditional on two named synthetic task families. Each independent unit is one task/base-model seed run; budgets and adaptation replicates are nested observations rather than additional independent samples.

Model, tasks, and calibration. Each run trains a two-layer decoder-only transformer with $d_{\text{model}} = 64$, four attention heads, MLP width 128, and zero dropout on a base task, freezes it, and adapts to a shifted task. Modular arithmetic changes an offset under modulus 31 and exposes eight q , v , o , and MLP-output sites. Associative recall changes a 16-key mapping and exposes ten sites, additionally including k projections. For site j , hooks estimate

$$\hat{C}_j = \frac{1}{N_j} \sum_{b,a} x_{j,b,a} x_{j,b,a}^\top, \quad \hat{G}_j = \frac{1}{B_j} \sum_b \sum_a \tilde{g}_{j,b,a} x_{j,b,a}^\top, \quad (14)$$

and form $\hat{M}_{j,\lambda} = -\hat{G}_j(\hat{C}_j + \lambda_j I)^{-1/2}$ with diagonal whitening and $\lambda_j = 0.01 \text{tr}(\hat{C}_j)/d_j$. Calibration uses eight batches of size 128, 4,096 permutation-null maxima per site, a 0.995 edge quantile, and 2,000 edge-uncertainty resamples. Rank sweeps use $\{0, 1, 2, 4, 8, 16\}$.

Prespecified comparison and controls. The primary allocation rule is soft spectral dimension under fixed update scale, compared with a deterministic uniform allocation constructed at the same realized trainable-parameter cost for every candidate allocation. Standard α/r and rsLoRA are separate sensitivity conditions. Adapter ini-

tialization, training batches, evaluation examples, and all declared random streams are shared across paired rules. Modular arithmetic is evaluated exhaustively. Each budget has three adaptation replicates; the six modular and seven associative-recall budgets are first averaged within a task-seed run. Five independent seeds per task are then averaged within task, and the confirmatory omnibus estimate weights the two task means equally. Inference uses 10,000 within-task run bootstrap resamples and an exact two-sided task-stratified sign-flip test over all 2^{10} patterns.

Confirmatory result. Figure 3 and Table 3 do not support an allocation advantage. The equal-task-weighted loss difference is +0.0146 (95% CI $[-0.0078, 0.0352]$, exact $p = 0.2402$), with 3/10 run wins and 0/2 task-family mean wins. The mean accuracy difference is -0.0039 . Associative recall has mean loss $\Delta = +0.0095$ and modular arithmetic +0.0198; both intervals include zero. The release contains 2,925 exact-cost candidate pairs, no realized-cost mismatch, no paired-seed disagreement, no divergent row, and zero metric spread among identical allocations, so the null result is not explained by the randomization and cost defects that invalidated the earlier analysis.

Scaling and task heterogeneity. Table 4 shows that the direction changes with the LoRA scaling rule. Under standard α/r , the exploratory mean is -0.0340 with 6/10 run wins and both task means negative, but the exact p value is 0.1973; rsLoRA is near zero overall. This is evidence of rank-scale sensitivity, not evidence for selecting the most favorable condition after observing outcomes.

Table 5: Sitewise association between spectral score and the near-best 0.1-gap useful-rank target under fixed update scale. Values are mean Spearman correlations over five independent runs, with SEM.

Task	Predictor	Mean ρ	SEM
Associative recall	Effective rank	-0.473	0.057
Associative recall	Soft dimension	-0.476	0.063
Modular arithmetic	Effective rank	+0.292	0.183
Modular arithmetic	Soft dimension	+0.304	0.179

Table 5 further separates module importance from rank demand. Effective rank and soft dimension correlate positively with useful rank in modular arithmetic but negatively in associative recall. A module can have a strong early-gradient spectrum yet achieve most of its attainable gain at low rank; a scalar score need not simultaneously encode whether a site matters and how many directions it needs.

Superseded analyses. The former six-cluster, 39-budget “33/39 wins” summary and its one-seed-per-task whitening ablation were generated before common-random-number and exact-cost controls. They are not publication

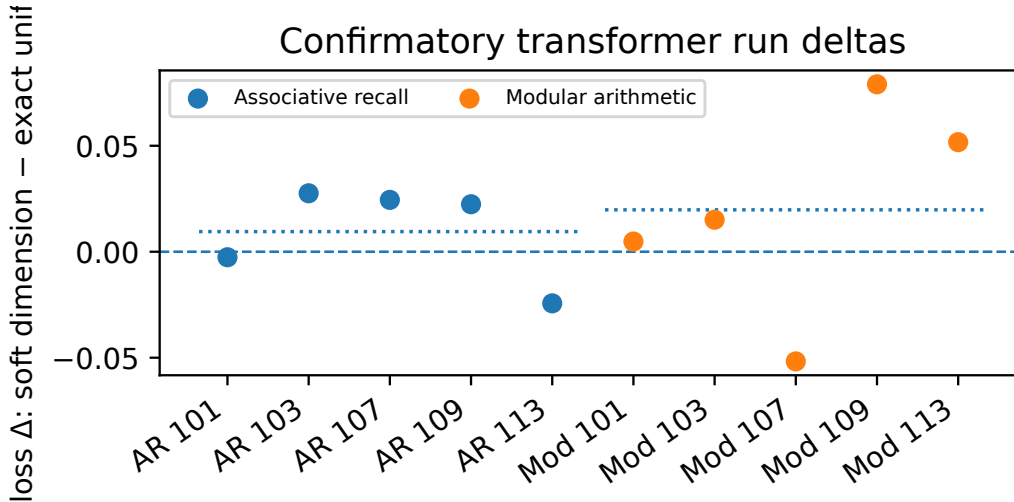


Figure 3: Confirmatory synthetic-transformer run effects. Each point is one independent task–seed run after averaging three adaptation replicates and the prespecified budgets. The outcome is final-validation-loss Δ = soft dimension minus its exact-cost uniform comparator, so positive values favor uniform allocation. Dotted segments are task means; the dashed line is zero.

Table 3: Prespecified fixed-update-scale transformer comparison. Positive loss Δ means the soft-dimension candidate is worse than exact-cost uniform allocation. The omnibus interval resamples independent runs within task and weights task means equally; p values are exact two-sided sign-flip tests. Task-specific rows are secondary and have five-run resolution.

Analysis	Runs	Mean loss Δ	95% CI	Exact p	Runs better	Mean acc. Δ
Omnibus (equal task weight)	10	+0.0146	[−0.0078, +0.0352]	0.2402	3/10	−0.0039
Associative recall	5	+0.0095	[−0.0102, +0.0253]	0.4375	2/5	−0.0015
Modular arithmetic	5	+0.0198	[−0.0197, +0.0553]	0.3750	1/5	−0.0064

Table 4: Scaling sensitivity for soft dimension versus exact-cost uniform allocation. Fixed update scale is confirmatory; standard α/r and rsLoRA are exploratory. Negative loss Δ favors the spectral candidate. None of the exact two-sided tests reaches 0.05.

Scaling	Role	Omnibus loss Δ	Exact p	Runs better	Assoc. Δ	Modular Δ
Fixed update scale	Primary	+0.0146	0.2402	3/10	+0.0095	+0.0198
Standard α/r	Exploratory	−0.0340	0.1973	6/10	−0.0293	−0.0388
rsLoRA	Exploratory	+0.0113	0.4609	3/10	−0.0004	+0.0231

evidence, are not inputs to any current table or figure, and are omitted from the claims. Whitening remains theoretically motivated, but its transformer allocation effect must be rerun under the corrected protocol.

7 Real GPT-2/Wikitext Validation

The synthetic experiments isolate the mechanism, but the practical motivation is LoRA allocation in pretrained transformers. We therefore add a small real-model validation using GPT-2 on Wikitext-2 [11, 12]. The experiment freezes a local pretrained GPT-2 checkpoint, calibrates early gradients on local Wikitext-2 blocks, and trains adapters for 200 steps under approximately matched trainable-parameter

budget caps. We compare uniform rank-4, gradient-norm allocation, and activation-whitened spectral effective-rank allocation. Candidate modules are GPT-2 Conv1D sites; all runs use the same local model, local dataset files, block size 128, batch size 4, eight calibration batches, and 32 evaluation batches. To isolate allocation from rank-dependent scale, this validation uses a fixed update scale (the released code’s `lora_alpha_scale=2.0`) rather than standard α/r scaling; scaling-rule ablations are future work.

Table 6 shows a limited positive real-model signal in the c_{attn}, c_{fc} setting. Spectral-effective allocation has the lowest mean final validation loss, wins four of five seeds, and uses the same mean trainable-parameter count as uniform rank-4. Its mean loss is lower than uniform by 0.0015 (paired-difference SEM 0.0013) and lower than gradient norm by 0.0475 (paired-difference SEM 0.0010). Given

Table 6: Real GPT-2/Wikitext-2 LoRA validation. Final validation loss is mean over seeds with SEM in parentheses. The c_{attn}, c_{fc} setting uses attention input projections and MLP input projections. The $+a_{proj}$ ablation additionally includes attention output projections.

Setting	Strategy	Final val. loss	PPL	Trainable params	Wins
c_{attn}, c_{fc}	spectral effective	3.4573 (0.0016)	31.73	331776	4/5
c_{attn}, c_{fc}	uniform rank 4	3.4587 (0.0017)	31.78	331776	1/5
c_{attn}, c_{fc}	gradient norm	3.5047 (0.0014)	33.27	330086	0/5
$+a_{proj}$	spectral effective	3.4530 (0.0026)	31.60	404992	0/3
$+a_{proj}$	uniform rank 4	3.4479 (0.0006)	31.44	405504	3/3
$+a_{proj}$	gradient norm	3.4847 (0.0014)	32.61	405248	0/3

only five seeds and the small separation from uniform, we treat the uniform comparison as descriptive rather than decisive. The allocation is structurally different from gradient norm: gradient norm spends rank on early high-norm attention modules, whereas spectral-effective allocation shifts capacity toward MLP input projections with larger activation-whitened effective rank.

The attention-output projection ablation is deliberately included as a boundary condition. When `attn.c.proj` modules are added, spectral-effective allocation still improves over gradient norm by 0.0317 mean validation loss, but uniform rank-4 wins all three seeds and is better by 0.0051 on average. The calibration reveals why: `attn.c.proj` modules are cheap per rank and have high whitened effective rank, so a naive cost-aware scalar spectral allocator over-allocates to that module family. This argues for group-constrained or module-family-normalized spectral allocation, not for dropping the spectral signal.

8 Rank Allocation Algorithm

The transformer experiments implement the following plug-in rule. For every candidate site j , compute the regularized whitened early gradient $\bar{M}_{j,\lambda}$. Let S_j be a scalar spectral score such as $r_{\text{eff}}(\bar{M}_{j,\lambda})$ or $d_{\phi, \hat{\tau}_j}(\bar{M}_{j,\lambda})$. A simple allocator chooses

$$r_j(\kappa) = \Pi_{\mathcal{R}_j}(\kappa S_j), \quad \sum_j c_j r_j(\kappa) \leq B, \quad (15)$$

where $c_j = d_{j,\text{in}} + d_{j,\text{out}}$ and κ is calibrated to the budget. A marginal-gain allocator instead estimates gains

$$\hat{b}_{j,i} = \frac{1}{2}\hat{\sigma}_{j,i}^2 \quad \text{or} \quad \frac{1}{2}(\hat{\sigma}_{j,i}^2 - \hat{\tau}_j^2)_+, \quad (16)$$

and includes directions with $\hat{b}_{j,i}/c_j > \lambda$. The population version of this rule follows directly from [Theorem 3.1](#): each singular direction reduces risk by $\frac{1}{2}\sigma_{j,i}^2$ at cost c_j .

The population marginal-gain interpretation is exact in the reduced-rank model, but the released transformer experiment does not validate the simple proportional soft-dimension rule as a universal allocator. A practical method must distinguish site importance from within-site rank demand, retain exact-cost uniform and gradient-based baselines, prespecify the LoRA scaling rule, and likely normalize

or constrain allocations by module family. We therefore present these equations as a theory-motivated design family rather than as a supported production algorithm.

9 Future Validation Targets

The next synthetic study should expand the number of task families and separate two predictions: a site’s attainable adaptation gain and the rank needed to recover that gain. Allocation rules should be preregistered with module-family constraints, exact realized-cost matching, fixed stochastic streams, and a single primary scaling convention; whitening must be replicated under these controls rather than inferred from the superseded ablation. Power should be based on independent task–seed runs, not nested budgets or adaptation replicates. A later open-model study should apply the same controls to several checkpoints and datasets, include stronger adaptive-rank baselines, and treat the present GPT-2 result only as pilot evidence.

10 Discussion

The strongest result is the reduced-rank connection between activation-weighted task spectra, early gradients, and useful rank. The five-seed matrix study supports that connection under both hard-knee and sample-limited regimes. The transformer study asks a distinct question and supplies a useful falsification: the primary scalar allocation rule does not beat an exact-cost uniform baseline, and its behavior depends on task family and rank scaling.

Limitations. The theorem is linear reduced-rank adaptation, not nonlinear transformer optimization. Stage4 is synthetic and uses 48 simulated layers per seed. The transformer allocation inference is conditional on only two prespecified task families and ten independent runs, so its intervals remain wide. The exploratory scaling results are not multiplicity-adjusted confirmatory tests. The real validation uses a small local GPT-2/Wikitext-2 setup, restricted target modules, short training, and fixed update scale; its uniform comparison is descriptive. Gradient

norm and exact-cost uniform allocation remain necessary baselines.

Claim boundary. The supported claim is that activation-whitened early-gradient spectra predict useful rank in the reduced-rank model and controlled spiked matrix simulations. The current evidence does not establish that a scalar spectral dimension should determine cross-module LoRA budgets. The corrected synthetic-transformer study is consistent with no benefit under fixed update scale and shows scale- and task-dependent directions. The GPT-2 result motivates further testing but does not override that confirmatory null result.

11 Conclusion

Useful low-rank adaptation is governed by activation-weighted task spectra in the reduced-rank population model, and early gradients reveal those spectra before adapter training. Multi-seed matrix simulations support prediction of practical useful-rank targets. The publication-grade synthetic-transformer experiment narrows the practical conclusion: soft spectral dimension does not outperform exact-cost uniform allocation under the prespecified fixed-scale condition, and the direction varies with scaling and task. Early-gradient spectra remain a principled diagnostic, but converting them into a reliable cross-module allocator requires additional modeling and broader validation.

Reproducibility Statement

All experiments write machine-readable metrics, frozen configurations, and provenance records. The corrected Stage4 table is imported from a five-seed release by `Paper/import_stage4_aggregate.py`. Corrected transformer tables are imported by `Paper/import_transformer_publication.py`, which verifies the release and archive checksums, rejects smoke or stale-plan releases, reconstructs ten primary run effects from raw exact-cost rows, checks paired stochastic streams and realized costs, and recomputes the confirmatory interval and exact test. The manuscript uses release `transformer_publication_20260622T101458Z`. Its full archive and frozen-plan SHA-256 values are recorded in `Paper/tables/transformer_publication/source.json`. `Paper/make_paper_artifacts.py` accepts only the imported post-CRN transformer tables for current figures and rows. Legacy pre-CRN transformer files remain solely for audit history and are excluded from manuscript generation. The real-model importer and corrected per-seed evidence retain input manifests, environment records, and SHA-256 checksums.

A Proofs and Additional Theory

Proof of Theorem 3.1. Using $y = (W_0 + \Delta_\star)x + \xi$ and $\mathbb{E}[\xi | x] = 0$,

$$\mathcal{L}(\Delta) - \mathcal{L}(\Delta_\star) = \frac{1}{2} \|(\Delta_\star - \Delta)C^{1/2}\|_F^2. \quad (17)$$

Let $M = \Delta C^{1/2}$. Because $C^{1/2}$ is invertible, $\text{rank}(M) = \text{rank}(\Delta)$. Hence the rank- r problem is

$$\min_{\text{rank}(M) \leq r} \frac{1}{2} \|M_\star - M\|_F^2, \quad (18)$$

whose solution is the truncated SVD.

Finite-sample perturbation. If $\widehat{M} = M_\star + E$ and $\|E\|_{\text{op}} \leq \varepsilon$, then $|\sigma_i(\widehat{M}) - \sigma_i(M_\star)| \leq \varepsilon$. Hence the number of empirical singular values above a threshold τ is trapped between the number of population singular values above $\tau + \varepsilon$ and the number above $\tau - \varepsilon$. This is the detectable-rank sandwich used to explain why hard thresholds require a spectral gap around the edge.

LoRA factorization bridge. In whitened coordinates write $M = sBA$, $s = \alpha/r$, and $F(M) = \frac{1}{2} \|M - M_\star\|_F^2$. The factorized global optimum equals the best rank- r approximation because BA represents any rank- r matrix. With zero- B initialization, one gradient step gives

$$M_1 = \eta s^2 M_\star A_0^\top A_0. \quad (19)$$

Thus the first LoRA step is a randomized spectral sketch of the task operator. Under factor gradient flow,

$$\dot{B} = -s(M - M_\star)A^\top, \quad \dot{A} = -sB^\top(M - M_\star), \quad (20)$$

$$\dot{M} = -s^2 [(M - M_\star)A^\top A \quad (21)$$

$$+ BB^\top(M - M_\star)]. \quad (22)$$

For one aligned mode with scalar factors a_i, b_i , $m_i = sa_i b_i$, and target singular value σ_i ,

$$\dot{m}_i = s^2(a_i^2 + b_i^2)(\sigma_i - m_i). \quad (23)$$

Only on the balanced positive branch $a_i^2 = b_i^2 = m_i/s$ does this reduce to $\dot{m}_i = 2sm_i(\sigma_i - m_i)$. Zero- B initialization is initially unbalanced: $m_i(0) = 0$ but $\dot{m}_i(0) = s^2 a_i(0)^2 \sigma_i$, consistent with the one-step formula above. The previously stated logistic equation is therefore a balanced special case, not the zero- B initialization dynamics.

B Reproducibility Details

The project generated archives for BBP, LoRA rank, alpha, merge, stage1b, stage2, stage3, Stage4, and transformer experiments. Each run writes CSV metrics and provenance files. The transformer publication design uses a two-layer decoder-only model with $d_{\text{model}} = 64$, four heads, MLP width 128, diagonal whitening with $\lambda = 0.01 \text{tr}(\widehat{C})/d$,

ranks $\{0, 1, 2, 4, 8, 16\}$, LoRA $\alpha = 16$, fixed-update-scale primary analysis, standard and rsLoRA sensitivities, five base seeds per task, and three adaptation replicates per budget. Stage4 fits use

$$\log_2 r_j^{\text{target}} = a + b \log_2(1 + s_j) + \epsilon_j, \quad (24)$$

where s_j is a spectral score. Main targets are $r_{\text{near}}(0.1)$, $r_{\text{near}}(0.2)$, $r_{\text{rec}}(0.7)$, $r_{\text{rec}}(0.8)$, $r_{\text{pen}}(0.2)$, and $r_{\text{pen}}(0.3)$.

Release commands. From `Code/`, `make test` runs the lightweight code tests. Stage4 smoke and publication releases use `run_stage4_smoke.sh` and `run_stage4.sh`; `make stage4-to-paper` imports only a validated five-seed release. Transformer smoke and publication releases use `run_transformer_publication_smoke.sh` and `run_transformer_publication.sh`. After validation, `make transformer-to-paper` independently imports the corrected ten-run aggregate. `make real-lora-to-paper` imports the GPT-2/Wikitext summaries, and `make paper` runs all three importers, regenerates current artifacts, compiles in `Paper/`, and copies the result to top-level `paper.pdf`.

Transformer inference resolution. The ten-run omnibus exact test enumerates 1,024 sign patterns under the task-stratified statistic and has nominal minimum nonzero two-sided resolution $2/1024 = 0.001953125$. Each five-run task-specific test is secondary and has minimum two-sided resolution $2/32 = 0.0625$. Budgets and adaptation replicates are averaged within their task-seed run and never counted as independent units.

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